

(1) Spin chain is described by $\hat{H}$

(2) $\hat{H}_T = \hat{H}_A + \hat{H}_{chain} + \hat{H}_B + \hat{H}_{AC} + \hat{H}_{BC}$

temperature

$\hat{\rho}_{chain}(0) = \frac{e^{-\beta \hat{H}_{chain}}}{Tr(\dots)}$

- $\hat{\rho}_T(0) = \hat{\rho}_A^{(0)} \otimes \hat{\rho}_{chain}^{(0)} \otimes \hat{\rho}_B(0) = \hat{\rho}_{AB}^{(0)} \otimes \hat{\rho}_{chain}^{(0)}$

$\hat{\rho}_{AB}(t) = Tr_{chain}\left\{ \hat{U}(t)\, \hat{\rho}_{AB}(0) \otimes \hat{\rho}_{chain}^{(0)}\, \hat{U}^\dagger(t) \right\} \equiv \mathcal{E}(\hat{\rho}_{AB}(0))$

$\hat{U}(t) = \exp[-i \hat{H}_T t]$

$\hat{\mathcal{E}}(\hat{\rho}_{AB}(0)) = \hat{\rho}_{AB}(t)$

$\hat{\rho}_{chain}^{(0)}$

* $\hat{\mathcal{E}}(\hat{\rho}_{AB}) = \sum_\alpha \hat{K}_\alpha(t)\, \hat{\rho}_{AB}(0)\, \hat{K}_\alpha^\dagger(t)$

(3) Hamiltonians (open boundary conditions)

$\hat{H}_A = \Delta_A \hat{\sigma}_A^z \quad ; \quad \hat{H}_B = \Delta_B \hat{\sigma}_B^z \quad ; \quad \delta = \Delta_A - \Delta_B$

$\hat{H}_{chain} =$
- Integrable
  - $\hat{H}_{XXZ}$
    - Gapped ferromag.
    - Gapless antiferromag.
  - $\hat{H}_{Ising}$
- Chaotic
  - $\hat{H}_{XXZ}$ + random fields
  - $\hat{H}_{XXZ}$ + local defect
  - $\hat{H}_{MFIM}$

$\hat{H}_{AC} = \lambda\left(\hat{\sigma}_A^+ \hat{\sigma}_C^- + \hat{\sigma}_A^- \hat{\sigma}_C^+\right)$ $\quad \lambda \ll$ couplings of the

$$\hat{H}_{AC} = \lambda\left(\hat{\sigma}_A^+ \hat{\sigma}_C^- + \hat{\sigma}_A^- \hat{\sigma}_C^+\right)$$

$$\hat{H}_{BC} = \lambda\left(\hat{\sigma}_B^+ \hat{\sigma}_C^- + \hat{\sigma}_B^+ \hat{\sigma}_C^-\right)$$

$\lambda \ll$ couplings of the $\hat{H}_{chain}$

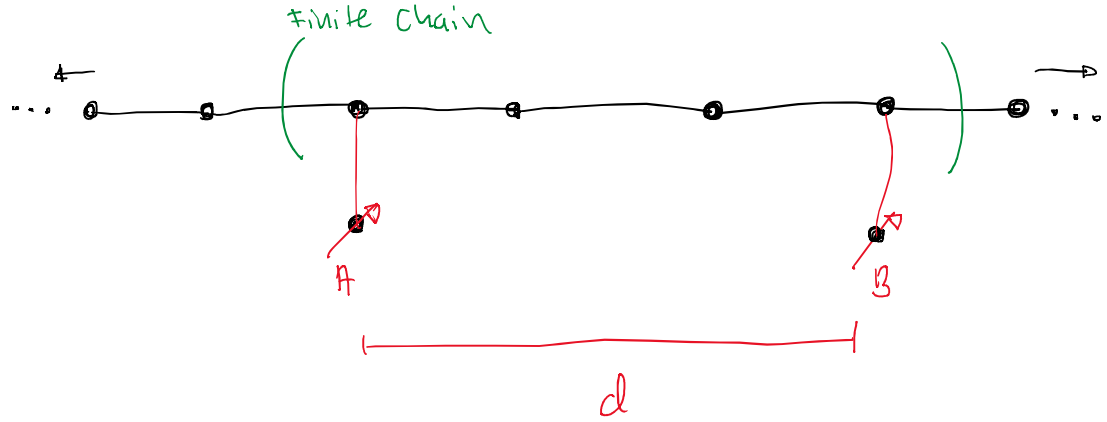
(*) Initial states

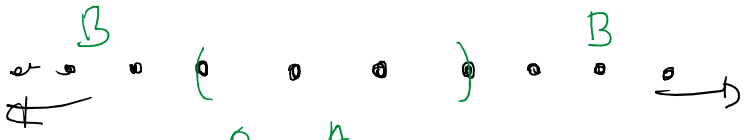
a) $\hat{\rho}_{AB}(0) = \hat{\rho}_A(0) \otimes \hat{\rho}_B(0)$

b) $\hat{\rho}_{chain}(0) = \dfrac{e^{-\beta \hat{H}_{chain}}}{Tr\left(e^{-\beta \hat{H}_{chain}}\right)}$

(*) How entanglement evolve in time?

(*) Limit of infinite size of the chain





transverse field ising model

ρ_A A

$S(\hat{\rho}_A)$; $\hat{\rho}_A = Tr_B(\hat{\rho}_T)$